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Ultrasound-assisted one-step extraction of safflower red and yellow pigments by natural ternary deep eutectic solvents: Extraction optimization and mechanism, separation and biological activities of pigments

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ABSTRACT

This study explores a green and sustainable approach based on natural ternary deep eutectic solvents (NATDES) and synergistic ultrasound-assisted extraction (UAE) of pigments from safflower florets. Choline chloride-ethylene glycol-acetamide (ChCl-EG-AM, 1:3:1) was successfully synthesized and characterized as the most effective NATDES for the one-step extraction of safflower red and yellow pigments. The optimal conditions for ChCl-EG-AM-UAE were as follows: solid-to-liquid ratio of 50.5 mg/mL, extraction temperature at 48 °C, extraction time of 21 min, and water content of 12.8 %. Moreover, this NATDES system could efficiently be recovered and reused in safflower pigments extraction. Safflower pigments also exhibit greater stability in ChCl-EG-AM under dark conditions at 4 °C. The AGREE, AGREEprep, and BAGI tools were used to evaluate this extraction process, suggestive of greenness. The Fourier transform infrared results showed that ChCl-EG-AM was mainly bonded by hydrogen bonding. Scanning electron microscope results revealed that ChCl-EG-AM disrupted the safflower florets' surface structure, facilitating the release of red and yellow pigments. Electrostatic potential, reduced density gradient, and molecular dynamic simulations were applied to reveal the extraction mechanism of ChCl-EG-AM-UAE, which demonstrated that hydrogen-bonding interactions were the key factor underlying its high performance. The isolated pigments (hydroxysafflor yellow A, anhydrosafflor yellow B, and carthamin) exhibit good antioxidant activity and cardioprotective effects in vitro. Overall, ultrasonic-assisted NATDES extraction is an eco-friendly method for the one-step recovery of safflower red and yellow pigments, which reduces resource waste and simplifies the process of extraction.

1. Introduction

Safflower (*Carthamus tinctorius* L.), belonging to the Asteraceae family, contains two major classes of bioactive pigments in its petals: red and yellow pigments (Zhang et al., 2024). The primary constituent of safflower red pigments is carthamin, while the yellow pigments are predominantly composed of hydroxysafflor yellow A (HSYA) and anhydrosafflor yellow B (AHSYB). Safflower pigments are valued not only for their good coloring properties but also for their potential health benefits, including cardio-cerebrovascular protection (Yu et al., 2024), hepatoprotection (Xue et al., 2024), and antioxidant properties (Zhang et al., 2023). To be specific, in the food industry, where consumers avoid synthetic pigments due to concerns over their potential health risks and environmental impacts, safflower pigments are widely used and accepted as natural pigments. For example, safflower pigments are commonly used to color beverages, confectionery, ice cream, pastries, and dishes (Popescu et al., 2022). In addition to the food industry, safflower pigments have an extensive spectrum of applications in the

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pharmaceutical field. Safflower yellow pigments are often used in traditional Chinese medicine to treat coronary heart disease, atherosclerosis, and other diseases (Bai et al., 2020 ; Gao et al., 2022).

Despite the widespread utilization, the extraction of safflower pigments remains challenging. Specifically, the conventional extraction techniques of safflower pigments, such as ethanol extraction, alkaline extraction, and acid precipitation, often encounter low yield, intricate procedures, prolonged extraction times, and the difficulty of achieving a one-step extraction of both red and yellow pigments (Wang, An, et al., 2025). More importantly, these extraction reagents (organic solvents, acids, and bases) are not only environmentally hazardous but also raise safety concerns due to their inherent toxicity. Therefore, these methods fail to comply with green chemistry principles. To address these issues, natural deep eutectic solvents (NADES) are considered as promising alternatives to conventional solvents. NADES are eutectic mixtures formed by mixing natural, biodegradable hydrogen-bond donors (HBDs) and hydrogen-bond acceptors (HBAs), resulting in mixtures with lower melting points than those of their individual components (Abbott et al., 2003). Their low toxicity, biodegradability, facile preparation, and modifiable physicochemical properties render them ideal candidates for sustainable extraction processes (Ran et al., 2025).

In recent years, NADES have been successfully applied to the extraction of various bioactive phytochemicals, demonstrating superior efficiency and environmental compatibility. In contrast to microwave-assisted extraction, ultrasound-assisted extraction (UAE) offers superior suitability for extracting thermosensitive compounds, such as the red pigment. UAE facilitates more precise control over the system temperature, thereby minimizing the risk of pigment degradation associated with improper microwave conditions (Bener et al., 2022; Cao et al., 2025). NADES and synergistic UAE have been widely used because of shorter time, higher extraction rate, lower energy consumption, etc. (Wang, Tian, et al., 2025). Although few studies have shown that NADES have a superior extraction efficiency for HSYA and AHSYB (Wu et al., 2025), no research to date has reported a one-step method for extracting safflower red and yellow pigments. Compared with the binary NADES system, ternary NADES (NATDES) seem to have a better extraction efficiency in phytochemical extraction (Yan et al., 2021). Therefore, we explored the directional optimization of NATDES-UAE for one-step extraction of safflower red and yellow pigments in this study. This work not only conforms to the principles of green chemistry but also meets the growing demand for natural pigments in industries such as food and medicine.

2. Materials and methods

2.1. Materials

Safflower florets were obtained from Shaanxi Sciendan Pharmaceutical Co., Ltd. (Shaanxi, China). Glucose and lactic acid were purchased from Tianjin Kemiou Chemical Reagent Co., Ltd. (Tianjin, China). Choline chloride (≥ 98 % purity), 1,4-butanediol, formamide, and sucrose were purchased from Sigma-Aldrich (Shanghai, China). 1,1-Diphenyl-2-picrylhydrazyl radical 2,2-diphenyl-1-(2,4,6-trinitrophenyl) hydrazyl (DPPH), 2,2′-azinobis-(3-ethylbenzthiazoline-6-sulphonate) (ABTS), L-proline, citric acid, tartaric acid, oxalic acid, betaine, DL-malic acid, malonic acid, and acetamide were obtained from Shanghai Yuanye Bio-Technology Co., Ltd. (Shanghai, China). Ehylene glycol and absolute ethanol were purchased from Tianjin Tianli Chemical Reagent Co., Ltd. (Tianjin, China). The AB-8 macroporous resin was purchased from Shaanxi Lebo Bio-Technology Co., Ltd. (Shaanxi, China). Ultrapure water produced by Milli-Q IQ7000 (Millipore Corp, USA) was used throughout the experiments.

2.2. Preparation and extraction of NADES

According to a specific molar ratio, various combinations of HBD and HBA were mixed in a beaker, maintaining a uniform water content of 20 %. The mixture was then heated and stirred in a constant temperature water bath at 80 °C for 2 h until a transparent and homogeneous liquid was achieved, which was subsequently allowed to cool naturally to 25 °C (Cai et al., 2021). The NADES tailoring and screening process was conducted in two rounds (Tables 1 and 2). Additionally, we selected comparison solvents, including water, methanol, L-proline-acetamide-water (L-Pro-AM-W) in a 1:1:2 mass ratio (Tong et al., 2021), and choline chloride: glucose: water (ChCl-GLU-W) in a 5:2:5 M ratio (Dai et al., 2013). Safflower florets were freeze-dried into a powder beforehand. For the extraction process, 0.3 g of dried safflower florets powder was thoroughly mixed with 2 mL of NADES using vortex agitation to ensure complete contact. The mixture was then extracted ultrasonically at 50 °C for 30 min (100 W, 40 kHz, $n = 3$). Following extraction, the samples were centrifuged (13,000 rpm, 10 min). The supernatant was carefully collected and transferred to an Eppendorf tube for subsequent analyses. The high-performance liquid chromatography (HPLC) diagrams of the reference substances and the ChCl-EG-AM extract (safflower florets extract using ChCl-EG-AM) are shown in Supplementary Fig. S1. The regression equations were shown in Supplementary Table S1.

Table 1
Types and molar ratios of the binary NADES in one round.

NO.	HBA	HBD	Short name and molar ratio
1	Betaine	Ethylene glycol	BE-EG 1:2
2	Betaine	Citric acid	BE-CA 1:1
3	Betaine	Glucose	BE-GLU 1:2
4	Betaine	Tartaric acid	BE-TA 1:1
5	Betaine	Lactic acid	BE-LA 1:1
6	Betaine	Propanedioic acid	BE-PA 1:1
7	Betaine	DL-malic acid	BE-DL-MA 1:1
8	Choline chloride	Ethylene glycol	ChCl-EG 1:3
9	Choline chloride	Citric acid	ChCl-CA 1:1
10	Choline chloride	Tartaric acid	ChCl-TA 1:1
11	Choline chloride	Lactic acid	ChCl-LA 1:1
12	Choline chloride	Propanedioic acid	ChCl-PA 1:2

Table 2
Types and molar ratios of the multicomponent NADES in the second round.

NO.	HBA	HBD 1	HBD 2	HBD 3	Short name and molar ratio
1	Choline chloride	Ethylene glycol	Acetamide	/	ChCl-EG-AM 1:3:1
2	Choline chloride	Ethylene glycol	Formamide	/	ChCl-EG-FM 1:3:1
3	Choline chloride	Ethylene glycol	Lactic acid	/	ChCl-EG-LA 1:3:1
4	Choline chloride	1,4-Butanediol	Lactic acid	/	ChCl-BDO-LA 1:3:1
5	Choline chloride	1,4-Butanediol	Acetamide	/	ChCl-BDO-AM 1:3:1
6	Choline chloride	Ethylene glycol	Oxalic acid	/	ChCl-EG-OA 1:3:1
7	Choline chloride	Ethylene glycol	Lactic acid	Acetamide	ChCl-EG-LA-AM 1:3:1:1
8	Choline chloride	1,4-Butanediol	Lactic acid	Acetamide	ChCl-BDO-LA-AM 1:3:1:1

2.3. Single-factor experiment design

To ensure the feasibility of the experiment, key factors that may affect the extraction efficiency of the target pigments were selected for single-factor experiment. Five independent variables were chosen: solid-to-liquid ratio (25, 50, 75, 100, and 150 mg/mL), extraction temperature (20, 30, 40, 50, and 60 °C), ultrasonic power (25, 50, 100, 150, and 200 W), extraction time (10, 20, 30, 40, and 50 min), water content (0, 10, 20, 30, and 40 %). The effect of varying each factor on the extraction rate of red and yellow pigments was investigated ($n = 3$).

2.4. Optimization of ChCl-EG-AM-UAE conditions

Based on the single-factor experiment results, four factors were selected as independent variables (Supplementary Table S2), with the red and yellow pigments extraction rates as the response values. Response surface methodology (RSM) was performed based on a Box-Behnken design (BBD). The interactions between the four factors were visualized to obtain the extraction process with optimum extraction rates for red and yellow pigments. Via each factor at three levels, a total of 29 trials were conducted, comprising 24 analytical factorization trials and five replicated trials ($n = 3$). Design Expert 13 software was used to analyze the data.

2.5. Fourier transform infrared (FT-IR) analysis

The Bruker Tensor 27 system (Bruker, Germany) was utilized to record FT-IR spectra. Samples were prepared using the potassium bromide tablet method. A certain amount of each solid sample (i.e., carthamin, HSYA, AHSYB, choline chloride, and acetamide) was mixed with dried potassium bromide powder, ground, and pressed into thin slices for FT-IR determination. An appropriate amount of the liquid samples, ChCl-EG-AM extract, and ethylene glycol were dropped into the potassium bromide tablet for FT-IR measurement. Scanning was taken in the range of 400–4000 cm^{-1} at 28 °C.

2.6. Scanning electron microscope (SEM) analysis

SEM (Tescan Vega 3, China) was utilized to analyze the morphology of the raw material and the extraction residues of safflower florets with different solvents. Specifically, after extracting safflower florets using the ChCl-EG-AM or other solvents (ChCl-EG-LA 1:3:1, ChCl-EG 1:3, methanol, water) in optimal extraction conditions, the residue was filtered and naturally air-dried to complete the sample preparation. A suitable amount of the sample was placed on a carrier stage and coated with a gold-palladium layer to enhance conductivity. The prepared sample was then positioned in the SEM chamber and imaged at an accelerating voltage of 5 kV using an electron beam to capture detailed images.

2.7. Separation and purification of carthamin, HSYA, and AHSYB

The appropriate amount of AB-8 macroporous resin was completely immersed in a beaker containing absolute ethanol, soaked overnight, and wet-loaded onto the column. The treated ChCl-EG-AM extract was loaded onto the column and statically adsorbed for 2 h. This was followed by gradient elution using ultrapure water, 20 % ethanol, 40 % ethanol, and 60 % ethanol, with elution volumes of 300, 100, 100, and 200 mL, respectively. After elution, the content and purity of cathamin, HSYA, and AHSYB in the eluent fractions were determined. The eluent fraction containing unseparated HSYA and AHSYB was concentrated by rotary evaporation and then subjected to separation and purification using Sephadex LH-20. The contents of carthamin, HSYA, and AHSYB in the eluent were detected, and the recovery was calculated using the following formula (1):

$D\ (\%) = (\rho_2 V_2 / (\rho_0 V_1)) \times 100$ (1).

D-recovery percentage; ρ_0-mass concentration of the target substance in the extraction solution (mg/mL); ρ_2-mass concentration of the target substance in the desorption solution (mg/mL); V_1-volume of the sample solution (mL); V_2-volume of the desorbed liquid (mL).

2.8. Green evaluation

To assess the environmental sustainability of the developed extraction methods and sample preparation stages, the AGREE (Pena Pereira et al., 2020), AGREEprep (Wojnowski et al., 2022), and BAGI (Manousi et al., 2023) indicators were employed.

2.9. Stability test

Safflower pigments were extracted using NADES and other conventional solvents under the optimized extraction conditions. The resulting extracts were then placed in three different environments (4 °C in the dark, 25 °C in the dark, and 25 °C with alternating day and night conditions) to detect changes in the concentrations of the three pigments over time ($n = 3$).

2.10. Theoretical study

The structures of choline chloride, ethylene glycol, acetamide, ChCl-EG-AM, HSYA, AHSYB, and carthamin were optimized at B3LYP/6–311++g (d,p) level. The quantitative analysis of molecular surface was performed by the Gaussian09 (Revision A.02) suite of programs and Multiwfn 3.8 software (Lu & Chen, 2012a; Lu & Chen, 2012b). The weak interaction analysis was performed in Multiwfn 3.8 software and VMD 1.9.3 software. The VMD 1.9.3 program was used to render the color-mapped isosurface graphs of electrostatic potential (ESP) and reduced density gradient (RDG) function analysis (Humphrey et al., 1996). Molecular dynamic (MD) simulations were performed using Material Studio 8.0.

2.11. Determination of in vitro antioxidant activity

The DPPH radical scavenging assay was conducted following a previously reported method (Jia et al., 2015). A 0.2 mmol/L solution of DPPH was prepared by dissolving it in absolute ethanol. HSYA was prepared in absolute ethanol, while AHSYB and carthamin were prepared in methanol, all at concentrations ranging from 1.25 to 0.0098 mg/mL. Take 100 μL of the sample solution and 100 μL of the DPPH solution, and mix them in a 96-well plate ($n = 3$). The absorbance should be measured at 517 nm after a 30 min reaction period in the dark. Since carthamin has a strong absorption at 520 nm, its absorbance can obscure the detection of free radicals, leading to a calculated scavenging rate that deviates from the true value. The clearance of DPPH was calculated. Each experimental set was repeated three times, with vitamin C (VC) serving as the positive control.

ABTS and potassium persulfate solutions were prepared with concentrations of 7 and 2 mmol/L, respectively, using ultrapure water, and then mixed thoroughly. The ABTS reaction solution was allowed to stand at room temperature, protected from light, for 12 h. The ABTS working solution was adjusted using phosphate buffered saline (PBS) at pH 7.4 (an absorbance of 0.7 ± 0.02 at 734 nm) and stored at −20 °C for further use. Stock solutions of HSYA, AHSYB, and carthamin were prepared in methanol at concentration ranges of 0.3125 to 0.0012 mg/mL for each pigment. The 100 μL from the sample solutions were added to a 96-well plate ($n = 3$), followed by 100 μL of the ABTS radical solution. After thorough mixing, the plate was incubated at room temperature in the dark for 6 min, after which the absorbance at 734 nm was measured. Each experimental set was repeated three times, and the ABTS radical scavenging activity was calculated.[29] VC was used as the positive control.

2.12. Cell experiments

2.12.1. H9C2 cell culture and cytotoxicity assay

H9C2 cells were cultured in DMEM (CM10090, Beijing Zhongke Maichen, China) supplemented with 10 % FBS (22011–8612, Zhejiang Tianhang Biotechnology Co., Ltd., China), penicillin (100 U/mL), and streptomycin (100 μg/mL, Shanghai Beyotime Biotech Inc., China) for multiple generations. The cells were seeded into 96-well plates

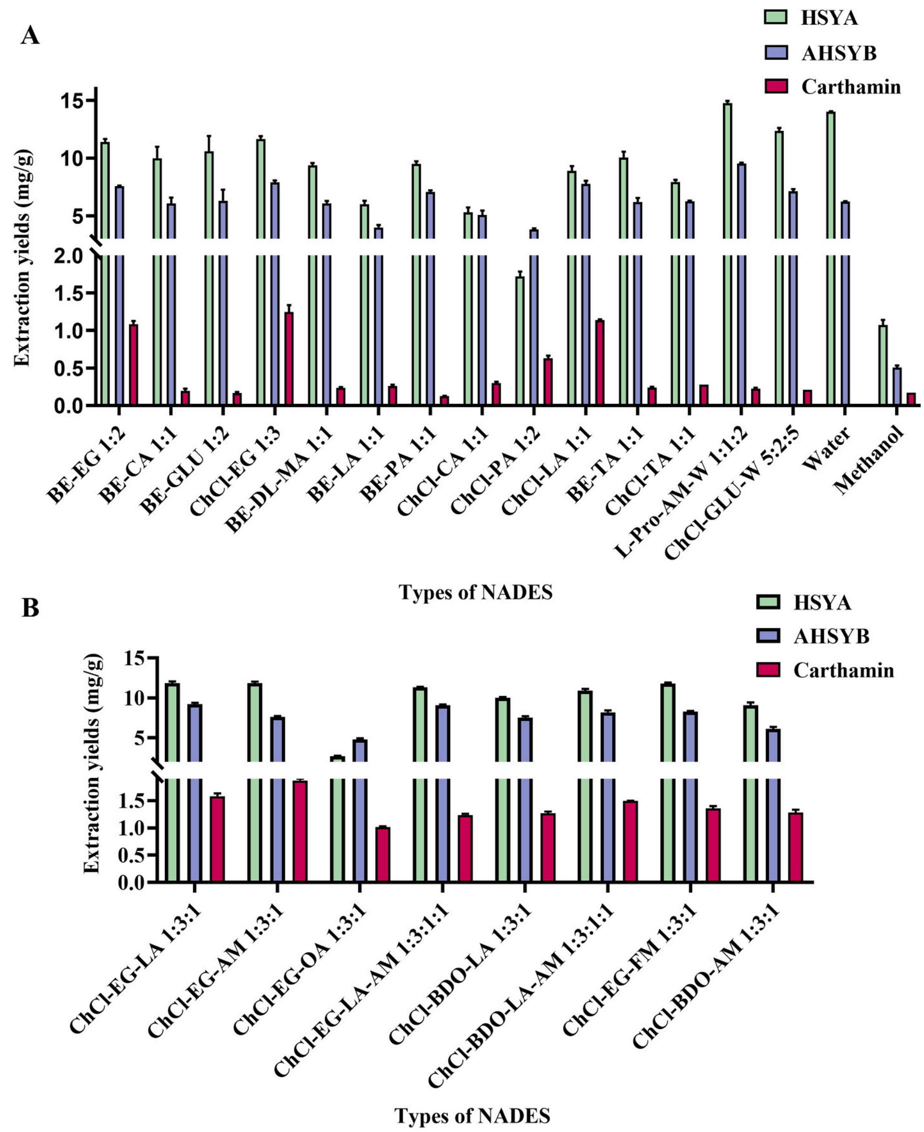

Fig. 1. (A) One round of the binary NADES extraction results; (B) Second round of the multicomponent NADES extraction results. L-Pro-AM-W (1,1:2), ChCl-GLU-W (5,2,5), water and methanol were selected as references, $n = 3$.

(2000–5000 cells/well) and allowed to adhere for 12 h (37 °C, 5 % CO_2). After two gentle PBS washes, medium was replaced with fresh DMEM containing graded concentrations of AHSYB, carthamin (10–200 μM) or HSYA (10–100 μM). Following 24 h exposure, cell viability was quantified with CCK-8 (CK04, Beijing DOJINDO, China) added for 2 h and absorbance read at 450 nm ($n = 4$).

2.12.2. Ang II-induced hypertrophy model, drug treatment, and immunofluorescence

To establish hypertrophy, cells were co-treated with 1 μM Ang II (HY-13948, MedChemExpress, USA) together with HSYA (10, 20, 50 μM), AHSYB (10, 20, 50 μM), carthamin (5, 10, 20, 50 μM) or captopril (10, 20 μM, positive control) for 24 h. Subsequently, cells were fixed (4 % paraformaldehyde, 15 min), permeabilised (0.1 % Triton X-100, 20 min), and blocked (5 % BSA, 1 h). α-Smooth muscle actin was labelled overnight at 4 °C with primary antibody; species-matched Alexa-Fluor secondary antibody was applied for 1 h at room temperature in the dark. Nuclei were counterstained with 4′,6-diamidino-2-phenylindole before imaging on a fluorescence microscope. Cell-surface area was quantified using Image-Pro Plus software (Media Cybernetics. USA).

2.13. Statistical analysis

GraphPad Prism 8.0 software (GraphPad Software, USA) was used for statistical analyses and graph generation. A single-factor analysis of variance (ANOVA) followed by Tukey's HSD test was used to determine significant differences. $P < 0.05$ was considered statistically significant.

3. Results and discussion

3.1. Extraction efficiency of NADES types

NADES typically exhibit high mass-transfer resistance, and their elevated viscosity significantly affects the extraction rate of safflower pigments (Patil et al., 2021). By adding an appropriate amount of water, we reduced the viscosity of the solvent system, which could enhance the extraction efficiency and lower the cost of NADES (Vilková et al., 2020). To investigate the effects of various NADES, we initially established the extraction conditions (a solid-to-liquid ratio of 150:1 mg/mL, water content of 20 %, extraction temperature at 50 °C, extraction power of 100 W, extraction time of 30 min, and ultrasonic frequency of 40 kHz). These findings indicate that different solvents exhibit varying abilities to

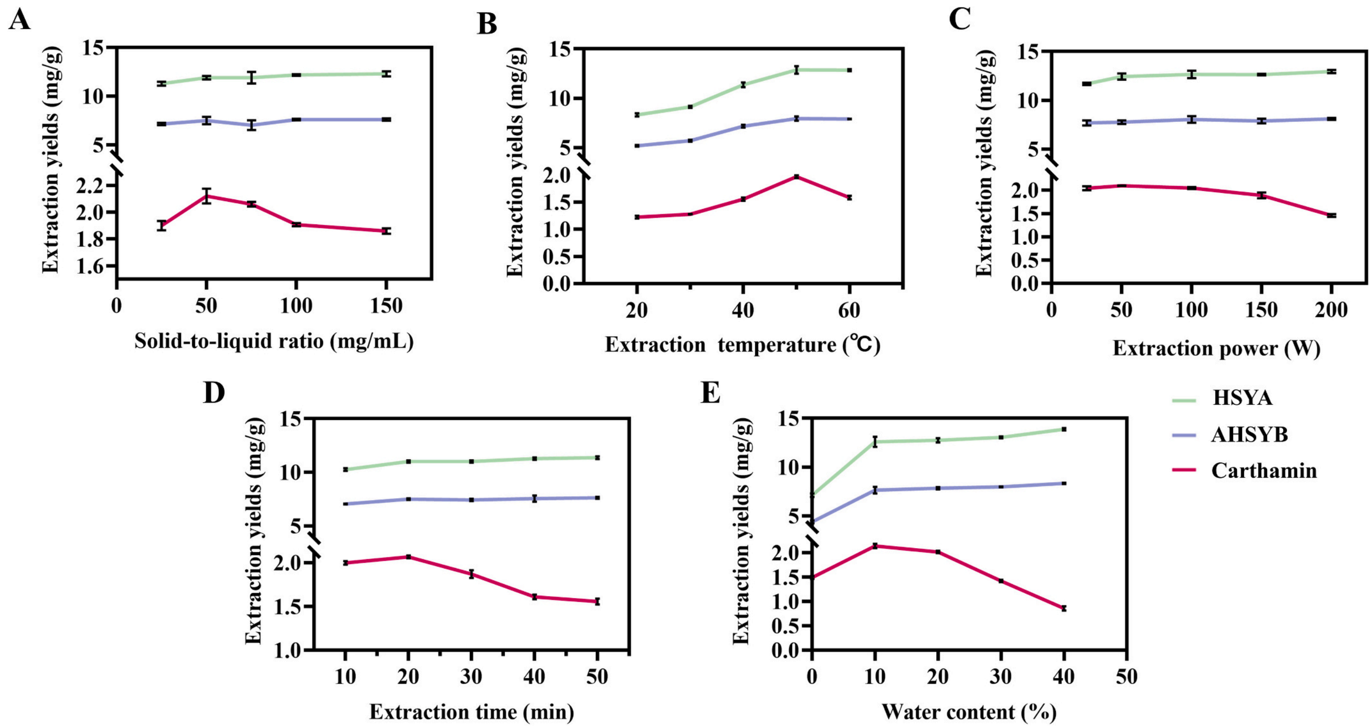

Fig. 2. Effects of single factors on red and yellow pigments contents: (A) solid-to-liquid ratio; (B) extraction temperature; (C) ultrasonic power; (D) extraction time, and (E) water content, n = 3. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

extract safflower pigments. During our initial screening using HPLC, we discovered that ChCl-EG (1:3) yielded the most effective extraction of safflower pigments. Subsequently, we designed and conducted a second round of screening based on these findings. In this round, we found that the combination of ChCl-EG-AM (1,3:1) exhibited excellent one-step extraction capability for both safflower red and yellow pigments, with a more pronounced color intensity. The results of these experiments are shown in Fig. 1. This approach not only leverages the unique properties of NATDES, avoiding the use of highly hazardous chemicals employed in traditional methods, but also enables the one-step extraction of safflower red and yellow pigments facilitated by the hydrogen bond regulation capability of NATDES.

3.2. Single-factor experiment on ChCl-EG-AM-UAE conditions

The results of the single-factor experiment depicted in Fig. 2 illustrate the effects of various parameters of NATDES and UAE on the extraction efficiency of safflower red and yellow pigments. Fig. 2A shows that the solid-to-liquid ratio has a minimal effect on the extraction rate of the yellow pigments while significantly impacting the red pigments. This discrepancy may be attributed to the large conjugated system present in carthamin; variations in the solid-to-liquid ratio can substantially influence the interaction forces between the solvent and solute. At low solid-to-liquid ratios, the binding capacity among solvent molecules is stronger than that between the solvent and pigment molecules, leading to a decrease in mass transfer kinetics between the solvent and solute, which results in a lower extraction rate. However, within the range of 25 to 50 mg/mL, the yield increases with the solid-to-liquid ratio, as the solvent penetrates the cell walls sufficiently, enhancing the interaction between the solvent and pigment molecules and thereby increasing the extraction rate. Conversely, when the solid-to-liquid ratio is excessively high, the solvent may not fully infiltrate the safflower floret tissue, preventing the release of internal red pigments and resulting in a lower extraction rate.

Extraction temperature is another important factor that affects extraction efficiency, and its influence on red and yellow pigments follows a similar trend. Fig. 2B illustrates that the extraction rates of both carthamin, HSYA, and AHSYB gradually increased between 20 °C and 50 °C. This increase may be attributed to higher temperature enhancing the mass transfer and solubility of the pigments. However, when the temperature exceeded 50 °C, the extraction rates began to decline. This decrease may be due to the presence of multiple hydroxyl groups and glycosidic bonds in the molecular structure of both red and yellow pigments. Elevated temperature may compromise the structural integrity of safflower pigments, resulting in a reduced extraction rate (Fan et al., 2011; Liu et al., 2022; Wang et al., 2024).

The impact of ultrasonic power was investigated on the extraction rate of safflower pigments (Fig. 2C). The findings indicated that ultrasonic power did not significantly influence the overall extraction rate of safflower pigments. Consequently, this factor was not included in the subsequent response surface experiments. As shown in the Supplementary Fig. S2, microwave does have a certain extraction ability for safflower pigments, but it cannot effectively extract both red and yellow pigments simultaneously. Moreover, with the increase of microwave power and extraction time, its ability to process red pigment significantly decreases compared with ultrasonic technology.

Fig. 2D illustrates that extraction time has a minimal impact on the extraction rates of the yellow pigments but significantly affects the extraction of red pigments. The extraction of carthamin initially increases with longer extraction times, peaking between 10 and 30 min. However, extraction times beyond this range lead to decreased efficiency, likely due to the thermal instability of the pigments and the leaching of other substances present in safflower florets (Ma et al., 2024).

In addition, the water content in NATDES significantly affects extraction efficiency, likely due to its impact on viscosity, pH, and other properties of NATDES. Fig. 2E illustrates that the optimal results for extracting red and yellow pigments were achieved with water contents ranging from 0 % to 20 %, with a more pronounced effect on the extraction of red pigments. Red pigments exhibit low polarity, and a moderate reduction in the viscosity of NATDES can enhance penetration. However, excessive water may compromise the NATDES hydrogen-

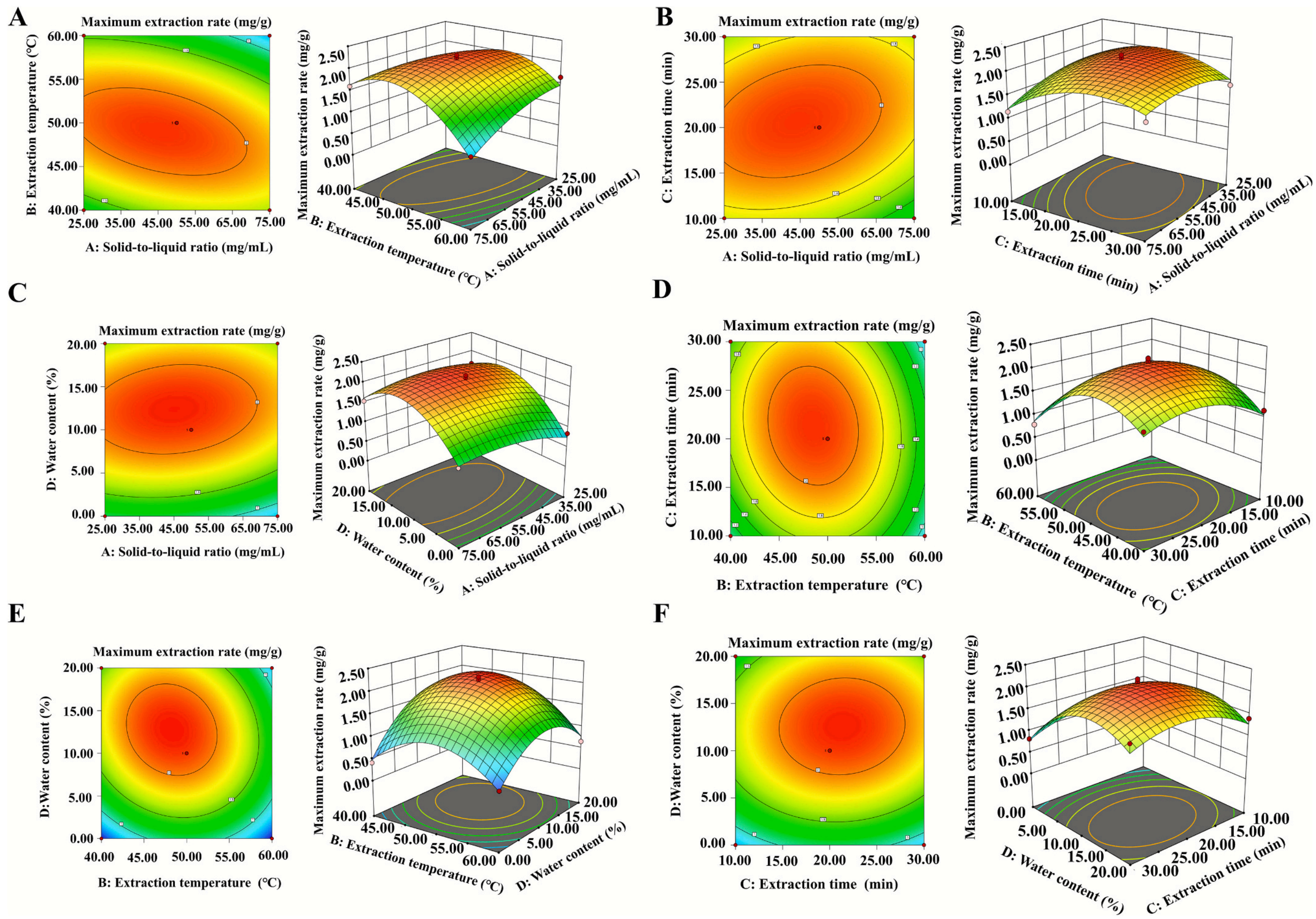

Fig. 3. Response surface plots and contour lines for the effects on the extraction rate of carthamin: (A) solid-to-liquid ratio and extraction temperature; (B) solid-to-liquid ratio and extraction time; (C) solid-to-liquid ratio and water content; (D) extraction time and extraction temperature; (E) extraction temperature and water content; (F) extraction time and water content.

bonding network and concurrently extract competing components, thereby reducing the pigments extraction efficiency (Liao et al., 2021). This underscores the importance of solvent composition in optimizing the extraction process.

3.3. RSM optimization of ChCl-EG-AM-UAE on red and yellow pigments

RSM optimization was employed to investigate the interactions among four key parameters (solid-to-liquid ratio (A), extraction temperature (B), extraction time (C), and water content (D)). The BBD consisted of four factors, three levels, and five centroids, and the three levels of each independent variable were expressed as −1, 0, and 1. The integrated evaluation value of the HSYA, AHSYB, and carthamin extraction rate was respectively taken as the evaluation index. The results of the HSYA and AHSYB response surface designs were shown in Supplementary Tables S2 and S3. The results of the carthamin were shown in Supplementary Table S3. The contour plots (Fig. 3) revealed the interaction between these factors affecting carthamin extraction efficiency. The data in Supplementary Table S4 were subjected to second-order polynomial regression analysis and ANOVA to establish a multiple quadratic regression equation: Maximum extraction $= 2.12–0.0983 \times A-0.1842 \times B + 0.0883 \times C + 0.3325 \times D -0.2800 \times A \times B + 0.1500 \times A \times C + 0.0950 \times A \times D - 0.1100 \times B \times C - 0.1775 \times B \times D + 0.05 \times C \times D - 0.1899 \times A^2–0.6687 \times B^2–0.3874 \times C^2–0.6562 \times D^2$.

BBD was employed to conduct an ANOVA to evaluate the response surface regression model for the maximum extraction rate of carthamin. In Supplementary Table S4, the R^2 of the model is 0.9647, meaning that 96.47 % of the variation in the carthamin extraction rate can be attributed to the independent variables. The model's *F*-value is 27.32, with a *P*-value <0.0001, indicating the model's significance.

The *P*-value (0.1126) for the mismatch term indicates that the mismatch term is not significant and that the regression equation is well-fitted across the regression region. The difference between R^2 (0.9647) and the adjR^2 (0.9294) was less than 0.2, supporting the adequacy of the model. Furthermore, as illustrated in Supplementary Table S4, the effects of the factors on the carthamin extraction rate varied. Factors A, C, and the interaction term BD had a significant effect on the extraction rate ($P < 0.05$), while factors B, D, the interaction term AB, and the quadratic terms A^2-D^2 exhibited a highly significant effect on the carthamin extraction rate ($P < 0.01$). Other interaction terms (AC, AD, BD, and CD) were not significant ($P > 0.05$) regarding the carthamin extraction rate, which may be attributed to the absence of a significant synergistic effect among the factors within this range.

To investigate the effects of various variables and their interactions on the carthamin extraction rate, response surface plots and contour lines were generated, as illustrated in Fig. 3. The contour lines for variables AC, AD, BC, CD, AB, and BD were elliptical. In contrast, the 3D surface plots for the response surfaces of AC, AD, BC, and CD were relatively flat, indicating no significant interaction between these variables ($P > 0.05$). Conversely, the 3D surface plots for the response surfaces of AB and BD were steep, suggesting a significant interaction ($P < 0.05$). This finding aligns with the results of the feasibility of practical operation: the solid-to-liquid ratio was 50.5 mg/mL, the extraction temperature was 48 °C, the extraction time was 21 min, and the water

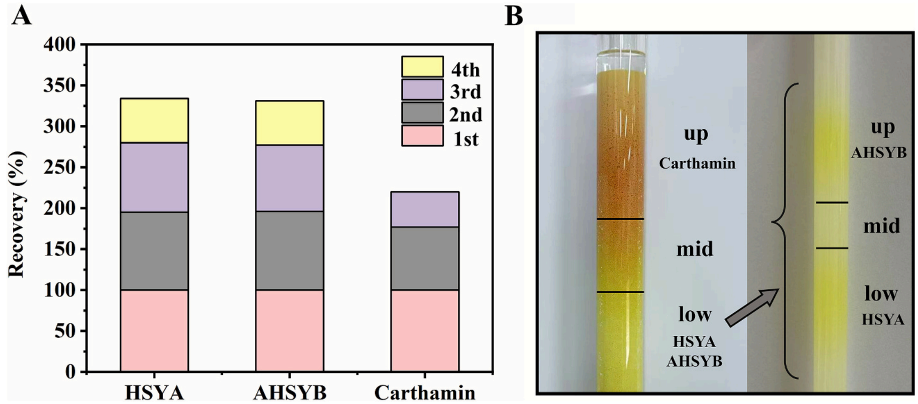

Fig. 4. (A) The extraction efficiency of solvent recovery; (B) The separation diagram of red and yellow pigments in AB-8 macroporous resin column, and HSYA and AHSYB in Sephadex LH-20 column. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

content was 12.8 %. Three sets of validation experiments were conducted in parallel following the optimized process described above. The mean yield of carthamin was found to be 2.1562 mg/g, which aligns closely with the predicted value of 2.1958 mg/g. The RSD was 2.83 % (< 3.0 %), indicating that the established model is both accurate and feasible and that the preferred process is stable and reliable. The response surface presented above reflects the results of the optimization for carthamin extraction. Based on the results of the response surface design for HSYA and AHSYB (Supplementary Table S5-S8), there is a discrepancy between the predicted and actual values for both compounds, indicating that the established HSYA and AHSYB methods were not reliable. We extracted HSYA and AHSYB using the optimal extraction process for carthamin and observed mean yields of 11.5838 mg/g and 6.6975 mg/g, respectively. These yields were within 13 % of the actual yield and fell within an acceptable range. Therefore, the established extraction process for carthamin can be effectively applied to one-step extract carthamin, HSYA, and AHSYB.

3.4. Infrared analysis

The FT-IR spectra (Supplementary Fig. S3A) illustrate the interactions involved in the synthesis of NATDES using choline chloride, ethylene glycol, and acetamide. In Supplementary Fig. S3A(d), it is evident that the sharp peak at 1500 cm^{-1}, when ChCl-EG-AM compared to the individual components of choline chloride, ethylene glycol, and acetamide, suggests that the carbonyl group underwent a Schiff base condensation reaction with the amino group. Additionally, the diminished or even absent absorption peak of the carbonyl group at 1700 cm^{-1} further confirms this condensation reaction. Concurrently, the -OH peak at 3300 cm^{-1} of the solvent compound exhibited a significant shift, indicating the formation of intermolecular hydrogen bonding and the successful establishment of the NATDES system.

Supplementary Fig. S3B illustrates that the infrared absorption peaks of HSYA, AHSYB, and carthamin are all present in the safflower florets extract obtained using ChCl-EG-AM. This indicates that ChCl-EG-AM has a highly effective extraction capability for these three pigments. Notably, compared to the monomers, the ChCl-EG-AM extract exhibits a blue shift in the -OH absorption peak at 3300 cm^{-1}, while the -C=O absorption peak at 1700 cm^{-1} shows a red shift. This further confirms that NATDES can efficiently extract the constituents of safflower florets.

3.5. SEM analysis

The SEM images of safflower pigments extracted using different solvents in optimal extraction conditions (ChCl-EG-AM, ChCl-EG-LA 1:3:1, ChCl-EG 1:3, methanol, water) revealed different effects on cellular structure and extraction efficiency (Supplementary Fig. S4). The structure of the original safflower florets showed a smooth surface and structural integrity under SEM, whereas all three NADES exhibited different degrees of disruption to the safflower florets and all were stronger than methanol and water, with ChCl-EG-AM causing the most severe structural disruption, generating visible pores and fissures in the cell wall, which contribute to better solvent interactions and hence improved extraction of pigments. The relative structural integrity of the safflower florets and the appearance of minor cracks on the surface after methanol and water extraction may be due to the weak ability of both to break down the cellular structure, which, in turn, exhibits a limited ability to extract the bioactivities. In conclusion, ChCl-EG-AM showed excellent cell wall disruption on safflower alone.

3.6. Separation and purification of carthamin, HSYA, and AHSYB

The crude extract was loaded onto the AB-8 macroporous resin column and adsorbed for 2 h. After washing with deionized water (300 mL), the eluent was concentrated to its initial volume via rotary evaporation, and the solvent was recovered for reuse. The results of pigments extraction using the recovered solvent are presented in Fig. 4A. Notably, carthamin extraction significantly declined by the third solvent recovery cycle and was undetectable in the fourth cycle. In contrast, HSYA and AHSYB retained their extraction capacity even after four cycles of solvent recovery. This may be due to factors such as the generation and accumulation of by-products, decreased acidity, and diminished hydrogen bonding in the solvent, all of which can impact the cycling performance of NATDES. The recycling of NATDES can not only reduce the generation of waste and improve the utilization efficiency of resources but also lower the cost of its application, which to a certain extent conforms to the principles of green chemistry (Sheldon & A., 2017).

AB-8 macroporous resin effectively separated carthamin from yellow pigments (HSYA, AHSYB). Further purification of yellow pigments was achieved using Sephadex LH-20, resolving HSYA and AHSYB. The purity of the purified carthamin was >95 %, with a yield of 66.34 %. For HSYA, two fractions were obtained: a high-purity fraction (> 97 % purity, 33.76 % yield) and a lower-purity fraction (> 85 % purity, 19.30 % yield). Similarly, for AHSYB, a high-purity fraction (> 97 % purity, 16.99 % yield) and a lower-purity fraction (> 85 % purity, 54.50 % yield) were obtained. AB-8 macroporous resin demonstrates good separation performance for red and yellow pigments, achieving a purity of up to 95 % (Fig. 4B). This phenomenon can be attributed to the distinct polarities of the target pigments. HSYA and AHSYB exhibit relatively high polarity due to their multiple hydroxyl groups and glycosyl groups. Consequently, when eluted with a low-concentration ethanol solution (polar), these yellow pigments are preferentially eluted. In contrast, carthamin possesses lower polarity owing to its larger benzene ring

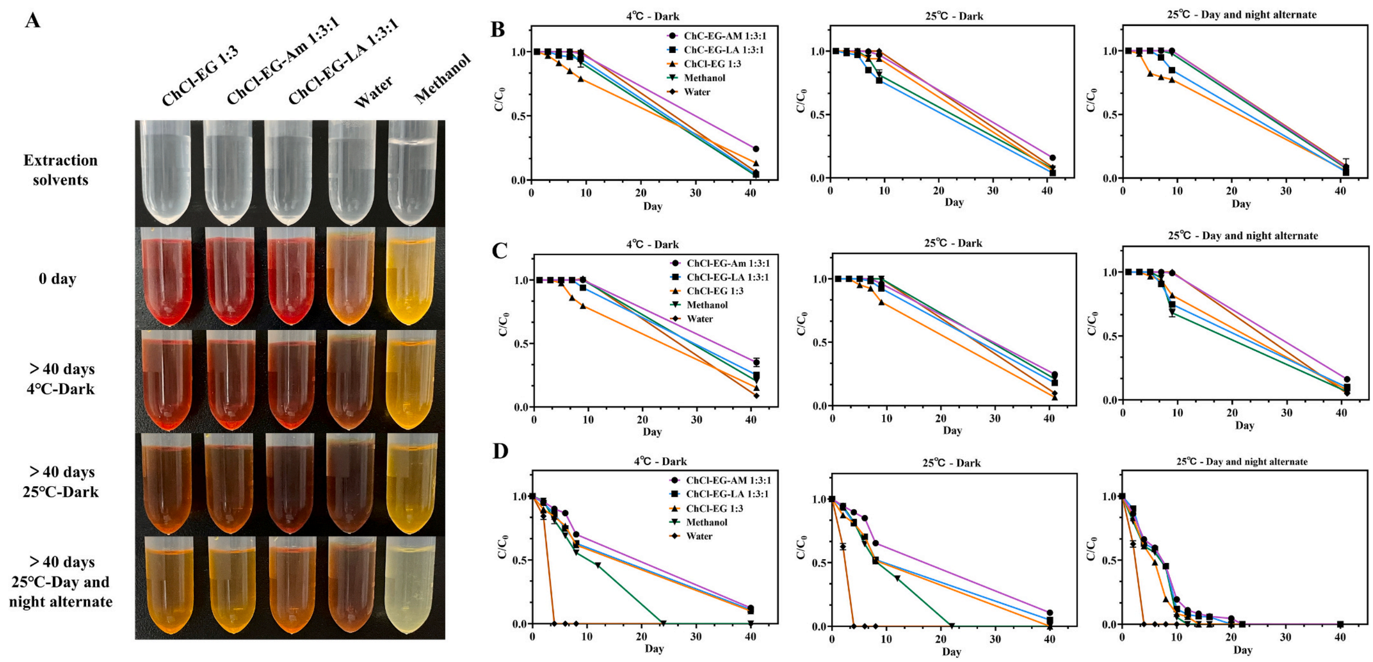

Fig. 5. (A) Color comparison: appearance of selected solvents, safflower extracts, and the extracts after >40 days of storage under different environmental conditions. The stability of HSYA (B), AHSYB (C), and carthamin (D) in the dark environment at 4 °C, in the dark environment at 25 °C, and alternating day and night at 25 °C, n = 3.

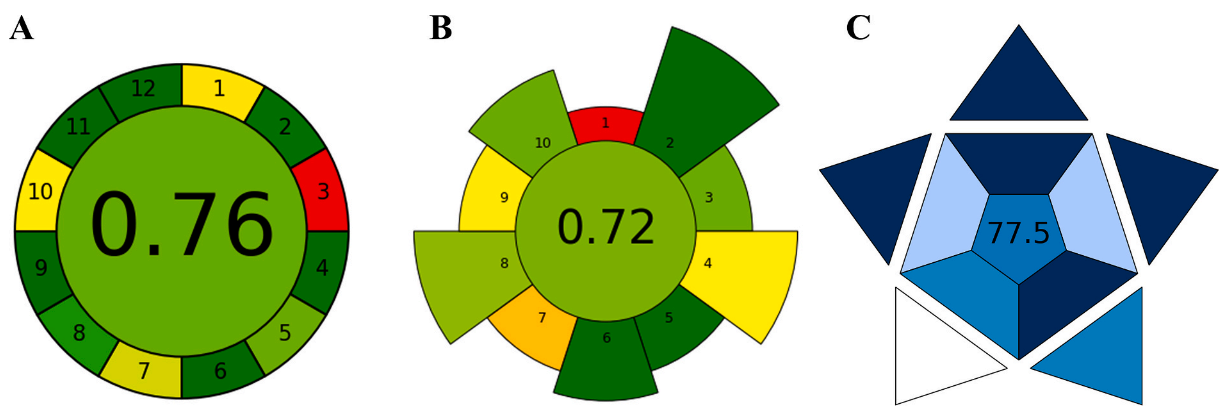

Fig. 6. Pictograms of AGREE (A), AGREEprep (B), and BAGI (C) generated for the ChCl-EG-AM-UAE method.

conjugated system until a higher-concentration ethanol solution (less polar) is applied. However, for HSYA and AHSYB, the separation performance of AB-8 macroporous resin is generally poor. Sephadex LH-20 was used to separate HSYA and AHSYB, and the results showed that the separation purity of both HSYA and AHSYB exceeded 85 %. The molecular weight of AHSYB is greater than that of HSYA, but HSYA may be expelled earlier due to its more relaxed molecular structure and larger effective volume within the gel. AHSYB may partially enter the gel pores due to cyclization or its more compact structure, resulting in an "apparently smaller" molecular size, which leads to an extended retention time. Therefore, Sephadex LH-20 has a good separation effect on HSYA and AHSYB, and the purity of HSYA and AHSYB will be higher after multiple purifications.

3.7. Stability of extracts in different environments for different days

Storage conditions are an important factor for maintaining the beneficial nutraceutical values of safflower pigments after extraction. The stability of HSYA, AHSYB, and carthamin in different extracts varied over time under different storage conditions, accompanied by observable color changes (Fig. 5A). Among the three pigments, HSYA and AHSYB exhibited greater stability, remaining detectable in the extraction solvent even after 40 days. In contrast, carthamin degraded more rapidly, particularly at 25 °C under alternating day-night conditions. Among the tested solvents, ChCl-EG-AM provided the highest stability for all three pigments. Optimal preservation was achieved under dark conditions at 4 °C, which was significantly better than storage in the dark at 25 °C or under alternating light/dark conditions at 25 °C (Fig. 5B-D). These results suggest that ChCl-EG-AM can serve as a multifunctioning medium for the efficient extraction and storage of safflower pigments.

3.8. Green assessment

As shown in Fig. 6A, the established method AGREE score was 0.76. The AGREEprep score was 0.72 (Fig. 6B). As a supplement to the green assessment tool, BAGI was also used to evaluate the applicability and practicality of the method, with a score of 77.5 after assessment

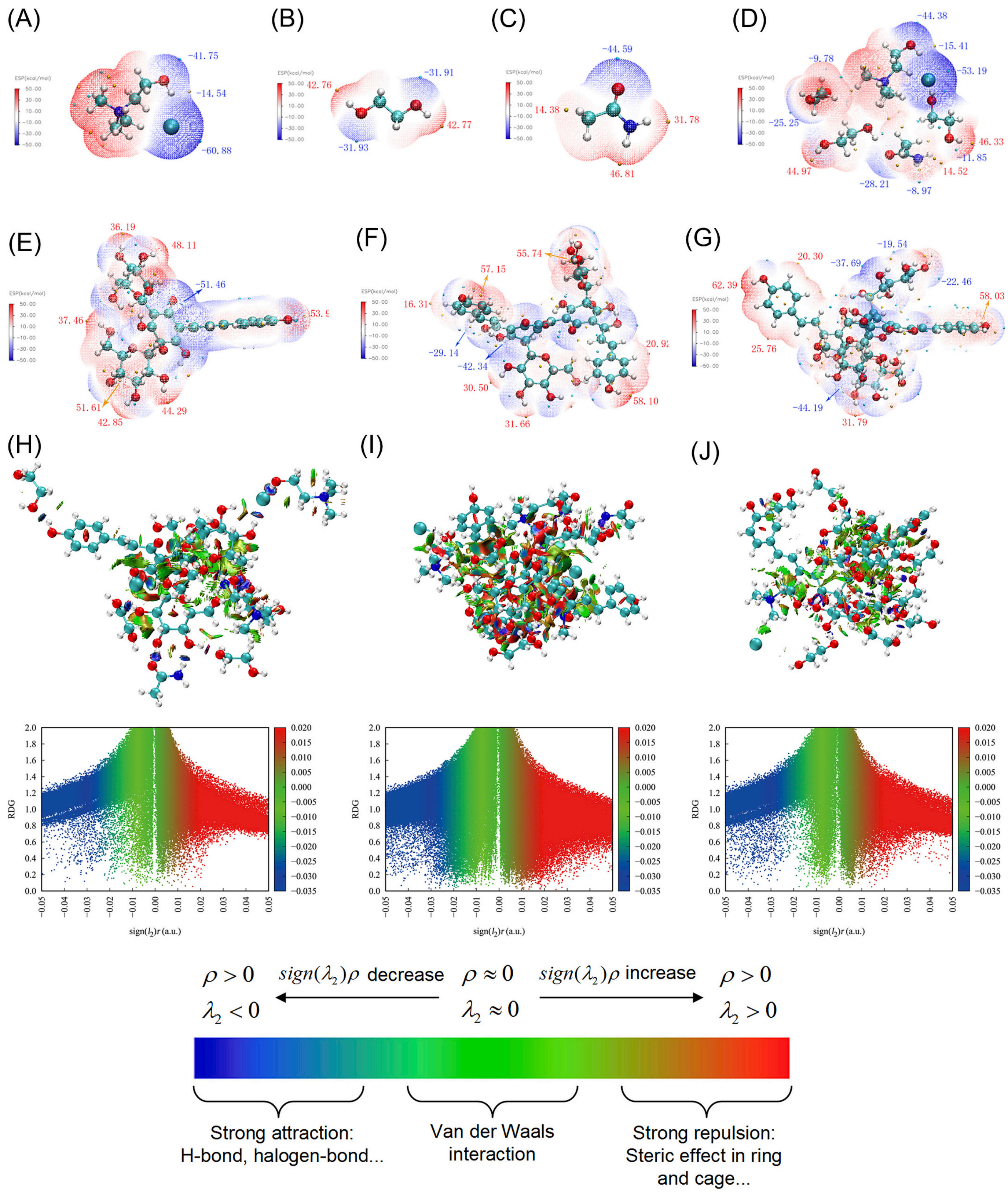

Fig. 7. ESP-mapped molecular vdW surface of (A) choline chloride, (B) ethylene glycol, (C) acetamide, (D) ChCl-EG-AM (1:3:1), (E) hydroxysafflor yellow A (HSYA), (F) anhydrosafflor yellow B (AHSYB), and (G) carthamin plotted on the 0.001 a.u. surface. The weak interaction analysis: (H) HSYA, ChCl-EG-AM and water; (I) AHSYB, ChCl-EG-AM and water; and (J) carthamin, ChCl-EG-AM and water. (C: cyan. H: white. O: red. N: blue. Cl: Cyan-Green. Maximum points: orange. Minimum points: cyan). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

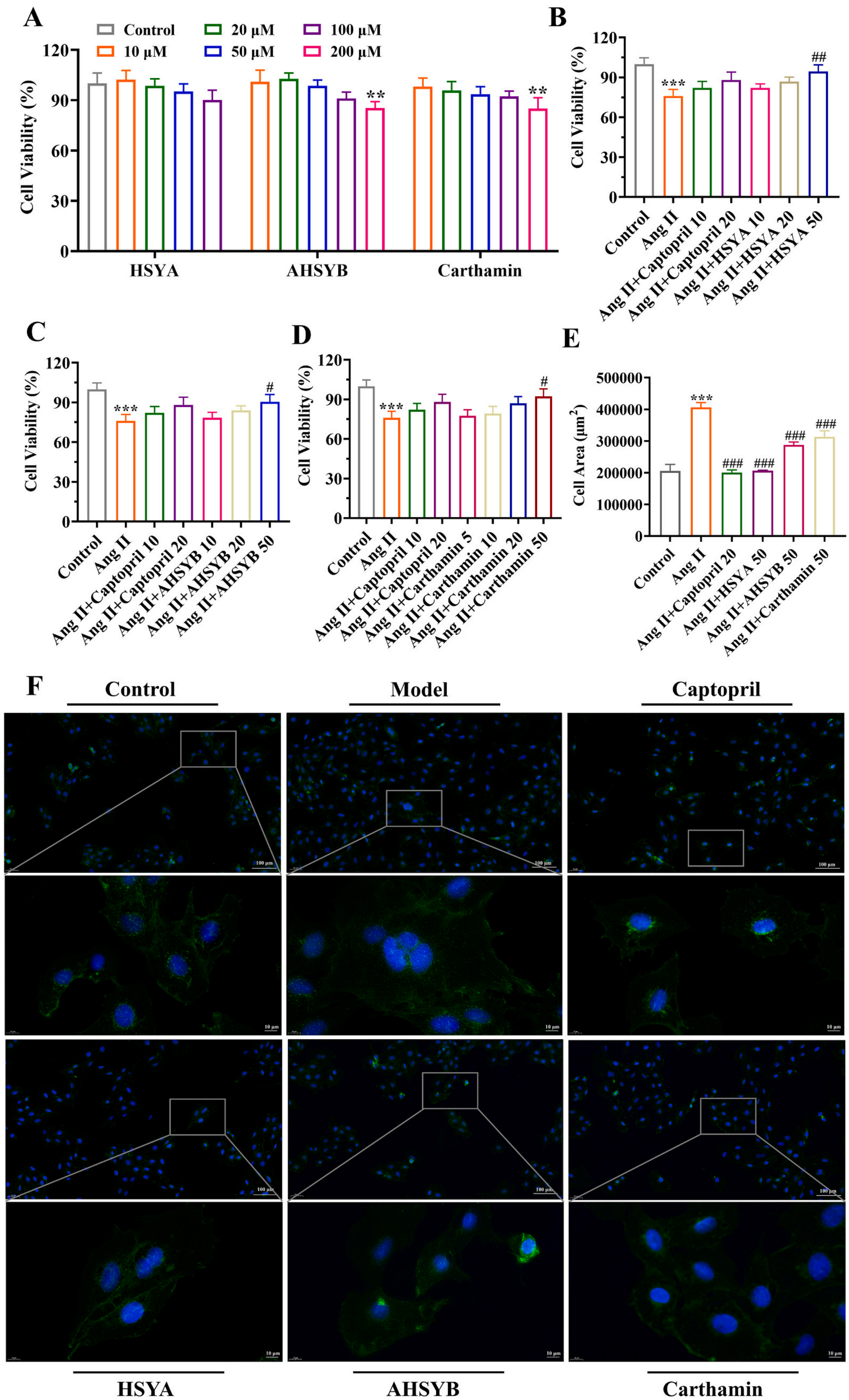

Fig. 8. (A) Survival of H9C2 cells after treatment with HSYA, AHSYB, and carthamin, $n = 4$; (B—D) Activity of H9C2 cells treated with HSYA, AHSYB, and carthamin respectively after induction with Ang II, n = 3; (E) The surface area of H9C2 cells in the different groups was measured with image magnification of 20×, n = 3; (F) Staining images of wheat germ agglutinin (WGA) by treating H9C2 cells in the different groups. All values are mean ± standard deviation. $^{**}P < 0.01$, $^{***}P < 0.001$ vs. control group, $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, $^{\#\#\#}P < 0.001$ vs. Ang II group.

(Fig. 6C). Based on these three pictograms, the proposed extraction process is in line with the current development trend of green chemistry. However, there is still room for improvement in some areas, such as the sustainability and renewability of materials, as well as the automation involved. Taken together, the developed method was considered a green extraction method.

3.9. Theoretical analysis

ESP, RDG, and MD simulations were applied to explain the mechanism of extraction performance of ChCl-EG-AM. As depicted in Fig. 7A-7C, choline chloride showed a global minimum negative potential of −60.88 kcal/mol near the chloride anion, and another local minimum negative potential of −41.75 kcal/mol near the oxygen atom of the hydroxyl group. Similarly, there are two global minimum negative potential of −31.93 kcal/mol and − 31.91 kcal/mol for ethylene glycol. As for acetamide, a global minimum negative potential of −44.59 kcal/mol was located near the oxygen atom of the carbonyl group. Fig. 7D revealed that intermolecular hydrogen bonds were formed in ChCl-EG-AM. Accordingly, many positive potential values can be found near the hydrogen of hydroxyl group in HSYA (Fig. 7E), AHSYB (Fig. 7F), and carthamin (Fig. 7G), ranging from 30.50 kcal/mol to 62.39 kcal/mol. Among these positive and negative ESP points, intermolecular interaction such as hydrogen bonding might occur. To further illustrate the weak interactions, RDG analysis was carried out. Fig. 7H-7J showed that strong attractions can be found between ChCl-EG-AM and safflower red and yellow pigments. In this case, hydrogen bonds were formed, which corroborate the earlier statements.

MD simulations were used to explore the extraction mechanism. The MD simulations considered 20 choline chloride, 60 ethylene glycol, 20 acetamide, 63 water molecules and 5 target pigment molecules (HSYA, carthamin, and AHSYB) in each box, respectively. All structures above were optimized by density functional theory calculations. As depicted in Supplementary Fig. S5A, HSYA molecules are thoroughly enclosed by ChCl-EG-AM and water molecules through hydrogen bonds, which consistent with ESP results. Similar conclusions can be drawn for carthamin, and AHSYB from Supplementary Fig. S5B and S5C. These theoretical results provided an explanation for the high extraction performance of ChCl-EG-AM.

3.10. In vitro antioxidant activities of safflower pigments

The DPPH radical scavenging activity of HSYA and AHSYB exhibited a noticeable increasing trend with higher concentrations, indicating that both compounds possess certain antioxidant capacity. However, their activities were lower than that of the positive control, VC, as illustrated in Supplementary Fig. S6A. The IC_{50} values for HSYA and AHSYB were 0.455 and 0.398 mg/mL, respectively.

The ABTS scavenging rates of HSYA, AHSYB, and carthamin exhibited a consistent increasing trend with higher concentrations, indicating that all of these pigments possess certain antioxidant capacity. However, their scavenging rates were lower than that of VC, as illustrated in Supplementary Fig. S6B. The IC_{50} values for HSYA, AHSYB, and carthamin were 0.06, 0.0014, and 0.0016 mg/mL, respectively.

3.11. In vitro cardioprotective activities of safflower pigments

Through the cell viability assay, HSYA, AHSYB, and carthamin did not exhibit significant toxicity to normal H9C2 cells at concentrations of 10–100 μM. When the concentrations of AHSYB and carthamin were 200 μM, they showed significant toxicity ($P < 0.05$) (Fig. 8A). On this basis, 50 μM of HSYA, AHSYB, and carthamin significantly enhanced the cell viability and reduced the surface area of hypertrophic H9C2 cells induced by angiotensin II (Ang II), and showed dose-dependent effects (Fig. 8A-8E). This might be due to the antioxidant properties of HSYA, AHSYB, and carthamin. Furthermore, compared with 50 μM AHSYB and carthamin, 50 μM HSYA produced the most pronounced improvements.

4. Conclusion

This study demonstrates that ChCl-EG-AM-UAE is an efficient and environmentally friendly method for one-step extraction of red and yellow pigments from safflower florets, simplifying the complexities associated with traditional extraction processes. The extraction process was evaluated using the green assessment tools AGREE, AGREEprep, and BAGI, confirming that it can be regarded as a green method. Under optimized conditions (solid-to-liquid ratio: 50.5 mg/mL, extraction temperature: 48 °C, extraction time: 21 min, water content: 12.8 %), ChCl-EG-AM achieved high extraction efficiency primarily through hydrogen-bonding interactions, as confirmed by FT-IR, ESP, RDG, and MD simulations. SEM revealed that this method disrupted the surface structure of safflower florets, facilitating the release of pigments. Moreover, safflower pigments exhibit greater stability in ChCl-EG-AM under dark conditions at 4 °C. The isolated pigments from safflower florets (i.e., HSYA, AHSYB, and carthamin) exhibited potent *in vitro* antioxidant and cardioprotective activities. While the NADES-UAE method offers a sustainable alternative to traditional organic solvents, future research should focus on promoting its transformation into an industrial application. To sum up, these findings contribute to the advancement of green extraction technology in the pharmaceutical field and nutritional health products, particularly in the utilization of natural pigments.

CRediT authorship contribution statement

Ximei Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Data curation. **Zhang Zhang:** Writing – review & editing, Methodology. **Xinyu Yang:** Writing – review & editing, Software, Resources. **Yixin Zhang:** Resources, Project administration. **Yuping Tang:** Supervision, Project administration. **Shijun Yue:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2025.147383.

Data availability

Data will be made available on request.

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